Sathwik Nellikoppa Basavaraja

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Summary

Software Engineer with 3+ years building scalable Python microservices on AWS. Experienced in building RAG and agentic AI proof-of-concepts, with a focus on rapid prototyping, AI integrations, and production-oriented backend systems.

Professional Experience

Binghamton University, School of Computing | Research Intern *Oct 2026 – Present*

- Researching machine learning methods for quantum computing by implementing and evaluating PyTorch-based models on quantum computing problems with Qiskit, analyzing experimental results, and documenting findings.

Accenture Solutions Pvt Ltd | Software Engineer *Dec 2022 – Jul 2024*

- Increased workload efficiency by 40% for enterprise-scale operations by designing and deploying scalable RESTful APIs in Python (FastAPI) and refactoring core backend data-processing pipelines.
- Reduced deployment time from 45 minutes to under 15 minutes across 3+ production services by engineering GitHub Actions CI/CD pipelines, parallelizing build stages, and automating PyTest and Jest testing gates.
- Enhanced processing efficiency by 45% for downstream React and TypeScript analytics dashboards by building data-intensive ETL pipelines that transformed and staged data from PostgreSQL into AWS S3.
- Boosted processing speed by 35% while resolving recurring production incidents by migrating legacy monoliths to cloud-native, EC2-hosted, Kubernetes-deployed Python microservices with Redis caching and Celery background tasks.

Accenture Solutions Pvt Ltd | Associate Software Engineer *Apr 2021 – Nov 2022*

- Accelerated core dashboard load speeds from 3.2s to 800ms by engineering high-throughput REST endpoints using Python (Flask), implementing server-side pagination, and tuning complex PostgreSQL database queries.
- Decreased database transaction spikes by 50% during peak traffic by auditing PostgreSQL execution plans, adding composite indexes, and refactoring inefficient SQLAlchemy/Django ORM queries.
- Lowered report generation latency by 30% for cross-functional stakeholders by engineering complex SQL queries across PostgreSQL and AWS S3 and delivering automated Power BI and Grafana dashboards.

The Valley Bootcamp | Software Engineer Intern *Jun 2019 – Aug 2019*

- Delivered user-friendly full-stack web applications with React, Python (Flask), JavaScript, and MongoDB, implementing authentication, search, filtering, and REST API integrations in an Agile team environment.

Projects

AI Knowledge Assistant (RAG / Agentic POC) | *Python, Streamlit, FAISS, BM25, GraphRAG, Ollama, Docker*

- Shipped a local-first, privacy-preserving retrieval-augmented generation (RAG) combining hybrid BM25 + FAISS retrieval, HyDE query expansion, and cross-encoder reranking with GraphRAG for single-hop reasoning, supporting pluggable local (Ollama) and cloud-hosted LLM inference, containerized with Docker for reproducible deployment.

Enterprise Knowledge Agent | *Microsoft Foundry, Foundry IQ, MCP, Python, RAG, Azure AI Search*

- Developed a knowledge-grounded AI agent using Microsoft Foundry and Foundry IQ to retrieve relevant information from connected knowledge sources, and integrated MCP for dynamic discovery and invocation of external tools.

Education

Binghamton University, State University of New York, Thomas J. Watson College of Engineering and Applied Science
Master of Science in Computer Science *Aug 2024 – May 2026*

Visvesvaraya Technological University, India
Bachelor of Engineering in Information Science and Engineering *Aug 2016 – Aug 2020*

Technical Skills

GenAI & Agentic: RAG, GraphRAG, Agentic Architectures, MCP, OpenAI API, FAISS, BM25, Prompt Engineering

Cloud & DevOps: AWS (EC2, S3, DynamoDB), Docker, Kubernetes, GitHub Actions, CI/CD Pipelines

Languages: Python, SQL, JavaScript, TypeScript, Shell Scripting, Java

Backend Frameworks: FastAPI, Django, Flask, SQLAlchemy, Django ORM, RESTful APIs, Microservices, PyTest

Databases, BI & Frontend: PostgreSQL, MongoDB, MySQL, Redis, Power BI, Grafana, React, Next.js, HTML5, CSS3

Developer Tools: Git, Postman, OpenAPI, GitHub Copilot, AI-Assisted Coding Workflows

Certifications: Microsoft Certified: Azure AI Apps and Agents Developer Associate, AWS Certified AI Practitioner